
PAPER CASE 03 / BIOTECH ATLAS

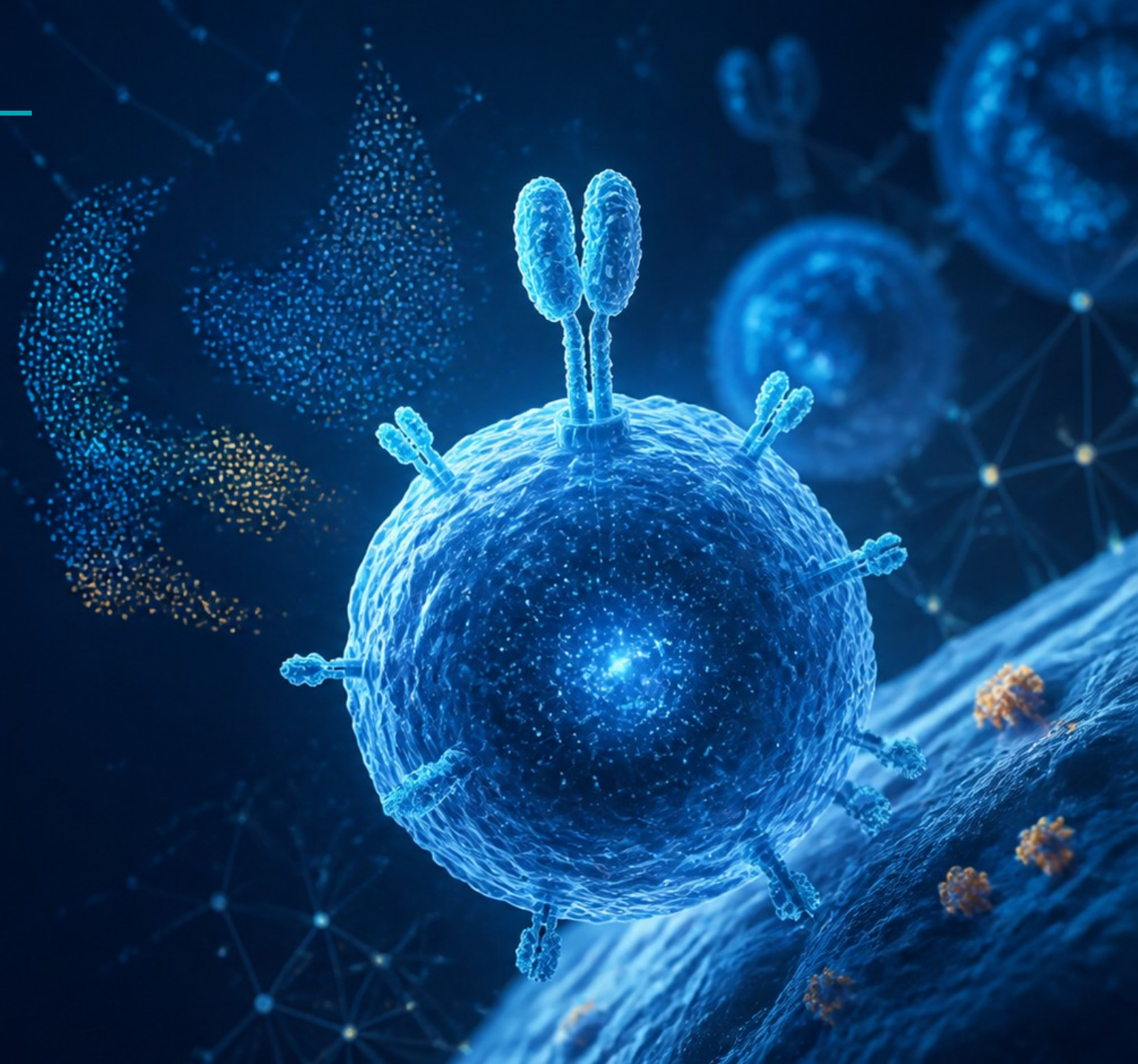
SINGLE-CELL CAR T ATLAS

**How a persistent clinical phenotype
becomes a cell-state hypothesis.**

Bai et al. · Nature 634 · 702–711 · 2024
82 patients · 695,819 analyzed cells
CONCEPTUAL COVER VISUAL

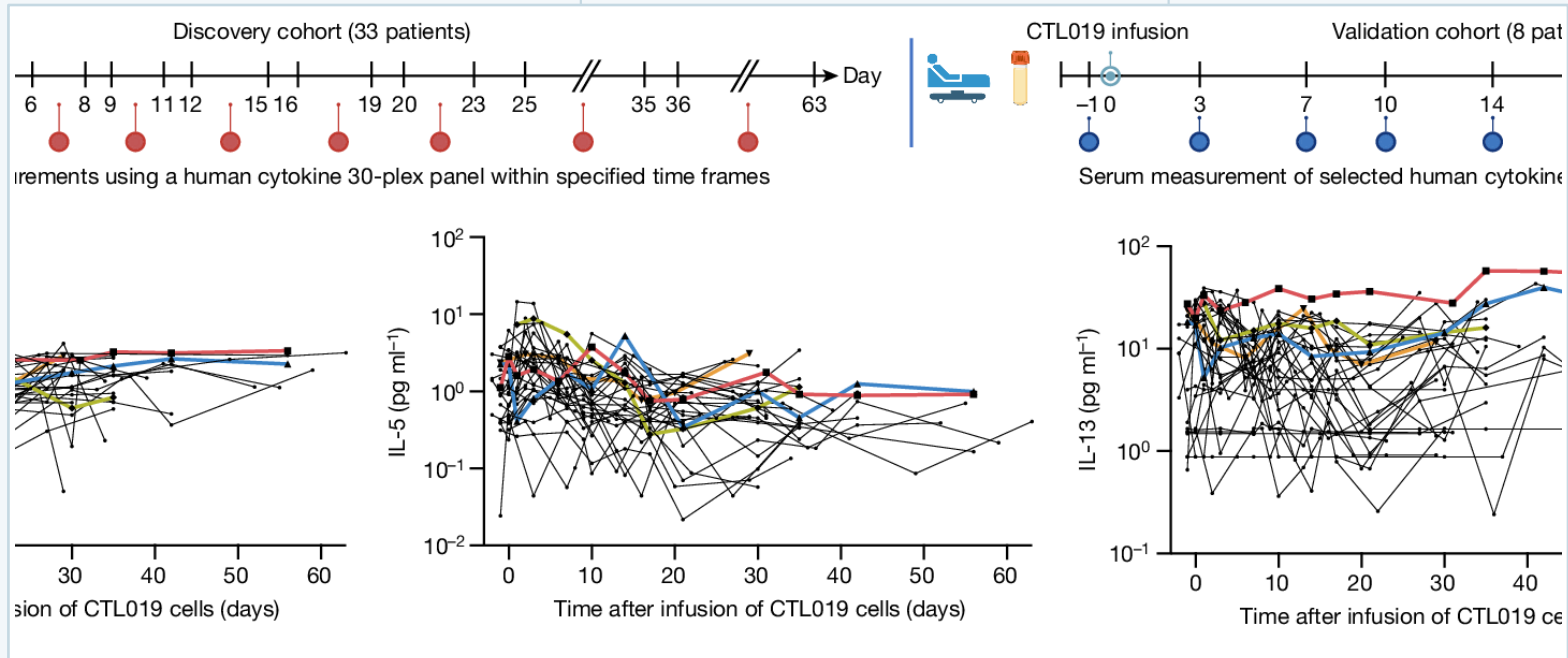
CLINICAL / CELLULAR / PRECLINICAL

DOI 10.1038/s41586-024-07762-w



The clinical signal is unusually long-lived

Before asking what the cell is, establish how long it remains clinically visible.



Source figure / Fig. 4a-b · longitudinal serum measurements

PERSISTENCE READOUT

8.4 years

median BCA-L duration

The phenotype is not a single measurement—it is a longitudinal trace.

OBSERVATION

long B-cell aplasia

QUESTION

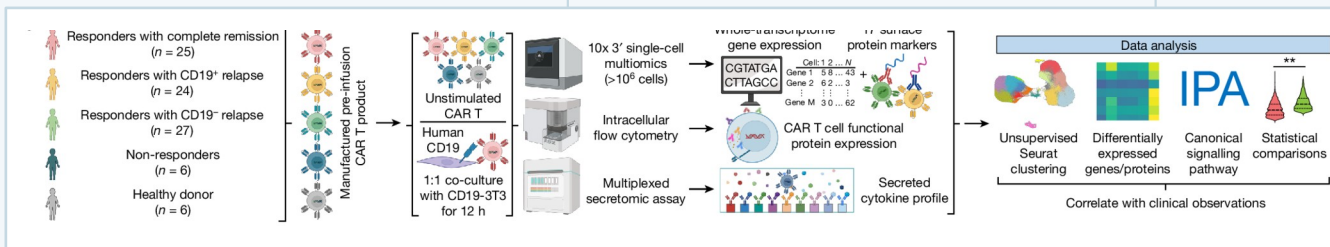
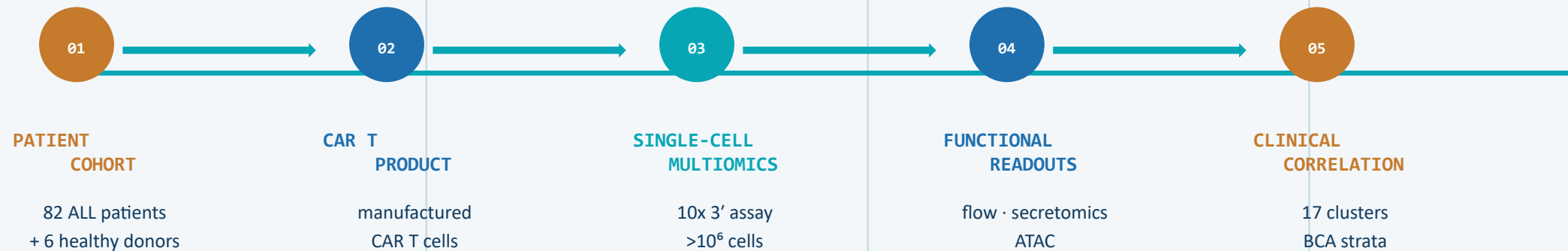
what sustains the state?

NEXT

single-cell atlas

One cohort becomes a multi-omic research system

The paper connects patient persistence to a measured cell-state atlas through five linked readout layers.



Source figure / Fig. 1a · cohort → measurement layers → data analysis

THE SYSTEM OUTPUT

695,819 analyzed cells

17 cell states · 5 BCA strata

Clinical observation becomes measurable state space.

695,819 cells resolve into 17 states

The UMAP is the map from which the later mechanistic claims depart.

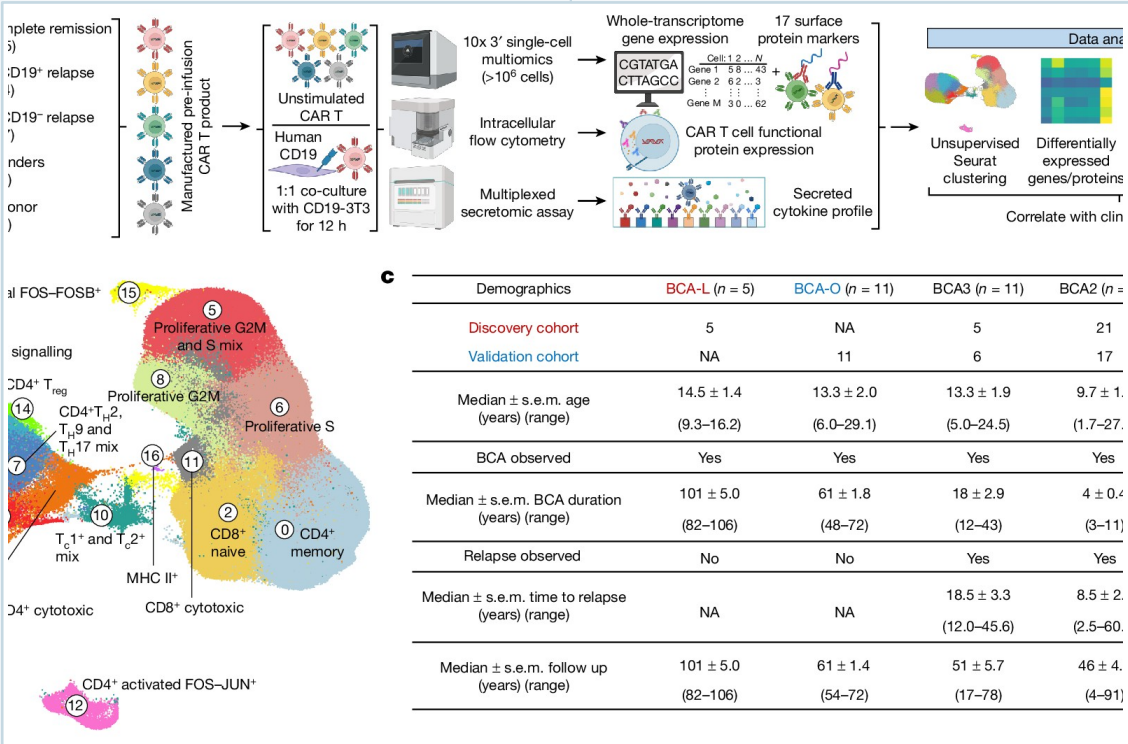


Fig. 1b / UMAP + cluster annotation

695,819
analyzed cells

17
cell states

ATLAS READOUT / FOUR STATE FAMILIES

NAIVE

CYTOTOXIC

REGULATORY

PROLIFERATIVE

Design principle: preserve the paper's cluster colors; use blue annotations to explain the map.

Persistence is a gradient, not a binary

BCA strata create the clinical axis reused across the paper.

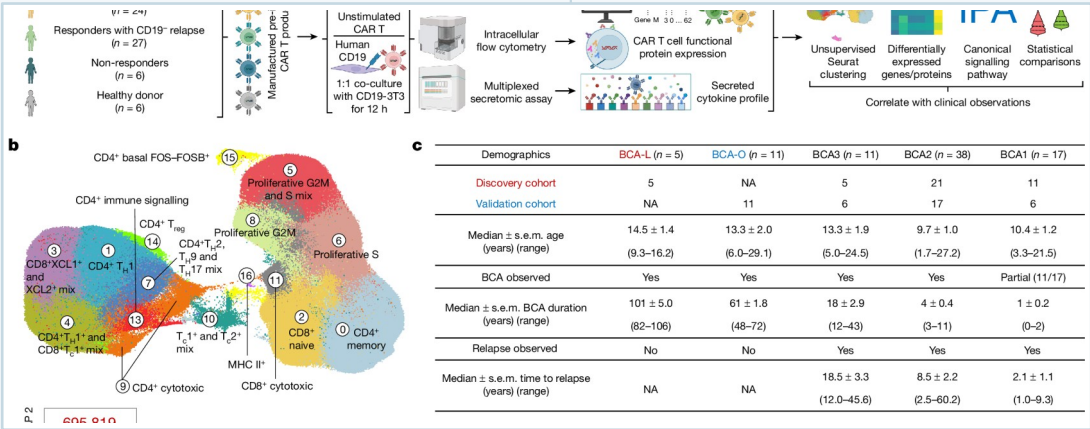


Fig. 1c / demographics and follow-up

PERSISTENCE GRADIENT



BCA-L / LONGEST

n = 5 · 101 months
no relapse observed

BCA1 / SHORTEST

n = 17 · 1 month
relapse observed

INTERPRETATION

Persistence is a continuum with a clinical gradient, not a yes/no label.

The long-remission state reads as type 2

Four orthogonal readouts point to the same biological state.

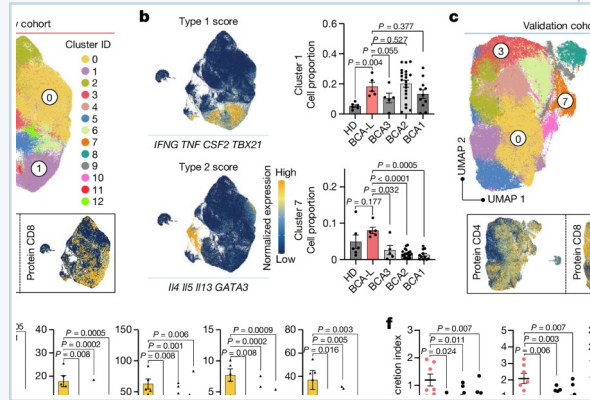


Fig. 2a-d / type 2 score

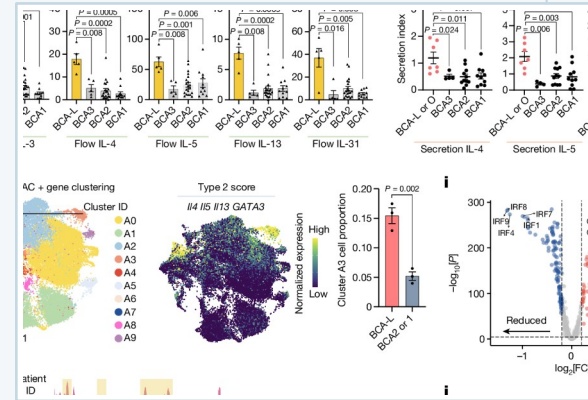


Fig. 2e-f / secretion

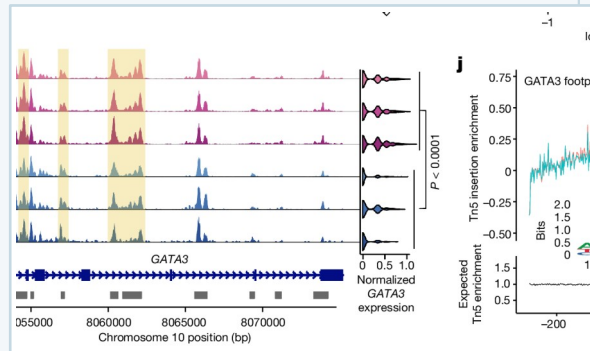


Fig. 2g-j / ATAC + GATA3

TYPE 2 SIGNATURE

IL4 · IL5 · IL13
GATA3

enriched in the long-remission group

PERTURBATION

GATA3 / STAT6

repeat stimulation

→ **tumour-cell lysis**

EVIDENCE RULE / CONVERGENCE

TRANSCRIPTOME → SECRETION → CHROMATIN

→ **one coherent type 2 state**

Cluster 2 is the pressure point

The paper moves from association to a candidate regulatory interaction network.

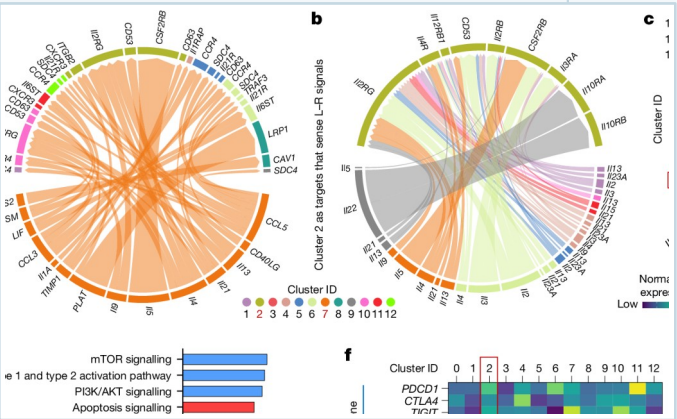


Fig. 3a-b / ligand-receptor network

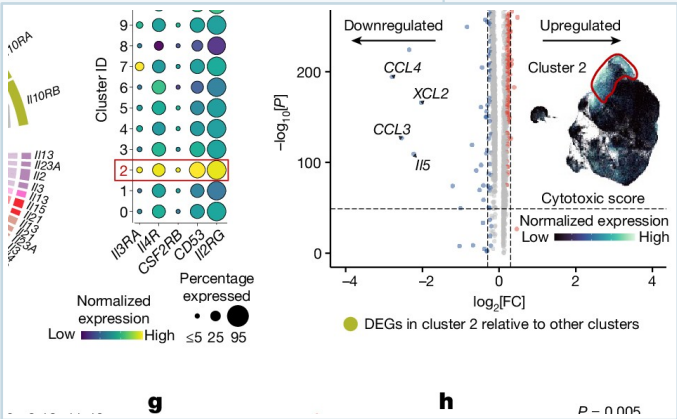


Fig. 3c-d / cluster 2 and DEGs

13.9%
cluster 2 proportion

STATE
type 2
signal
cluster 2
proliferation

CLAIM BOUNDARY

Candidate mechanism
not a complete causal proof

EDITABLE MECHANISM SUMMARY

Type 2 function → cluster 2 → dysfunctional subpopulation

The serum trace keeps the biology longitudinal

A state is more credible when it survives time, not just a snapshot.

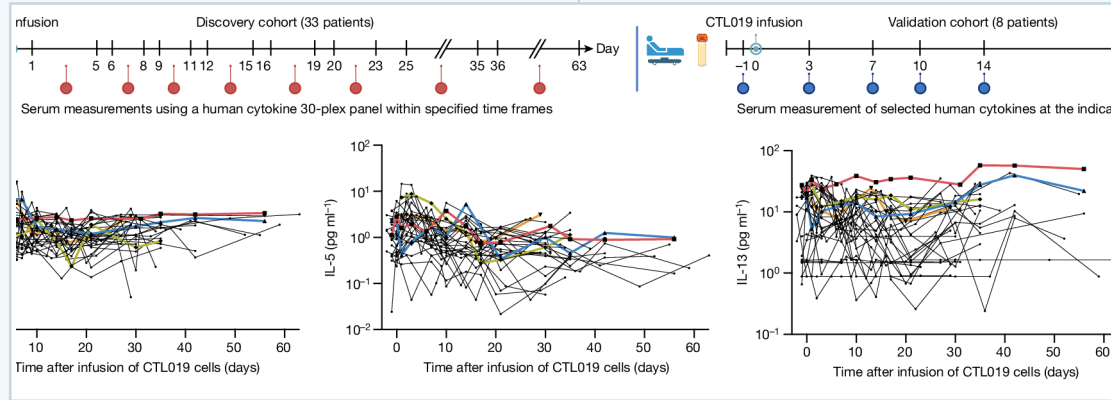


Fig. 4a-b / collection timeline + time series

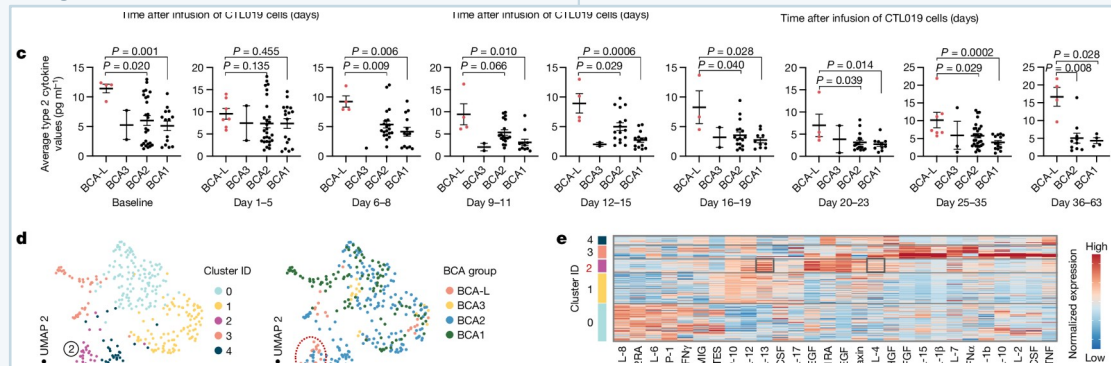


Fig. 4c-e / cytokine windows + heatmap

345
measurements

30
cytokines

33
patients

TYPE 2 WINDOW

IL-4 · IL-5 · IL-13
serial evidence across days 1–63
33 discovery + 8 validation patients

Type 2-high cells pass the recall test

The state is tested as function, not merely described as phenotype.

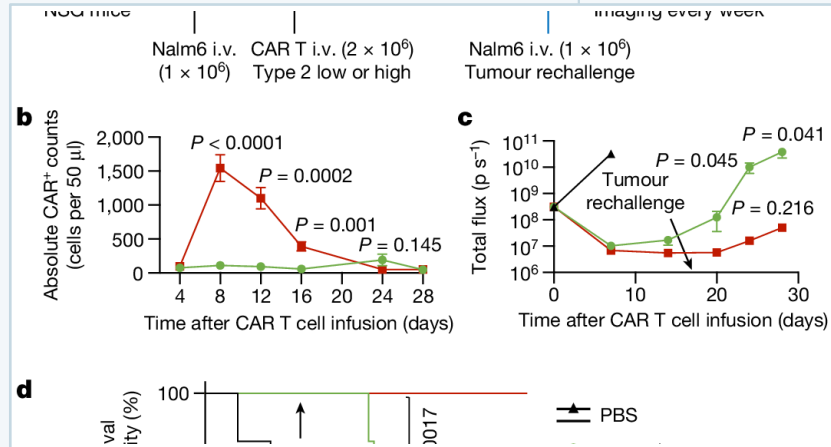


Fig. 5a-c / mouse sequence + expansion

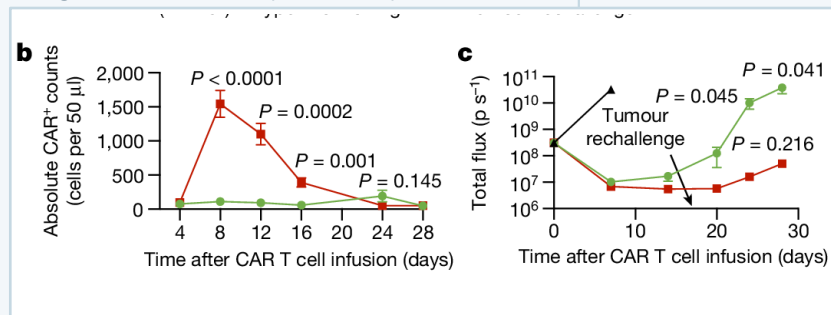


Fig. 5d / survival after rechallenge

IN VIVO SEQUENCE



READOUT

Type 2-high cells expand more and show stronger recall in the reported model; selected readouts n=5 mice.

EVIDENCE BOUNDARY

Preclinical model supports function; it does not establish clinical efficacy.

IL-4 turns the state into a design lever

The final figure converts a cell-state hypothesis into a manufacturing strategy.

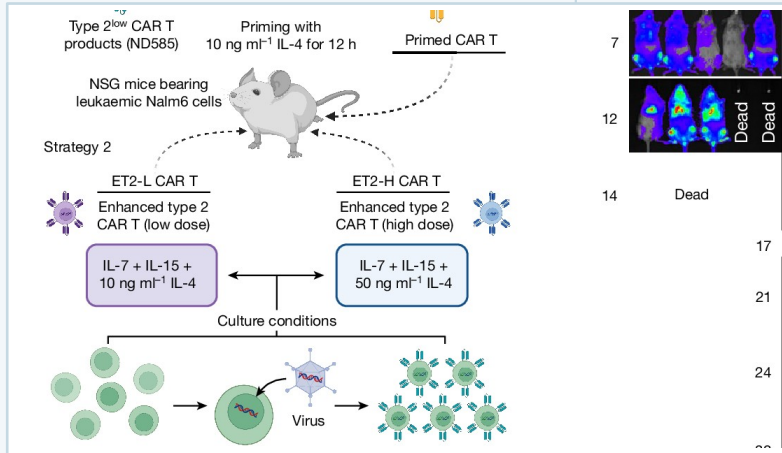


Fig. 6a / priming and ET2-L/H strategies

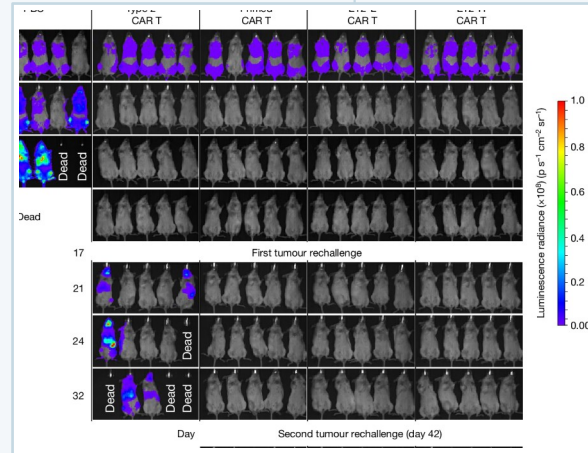


Fig. 6b / in vivo imaging

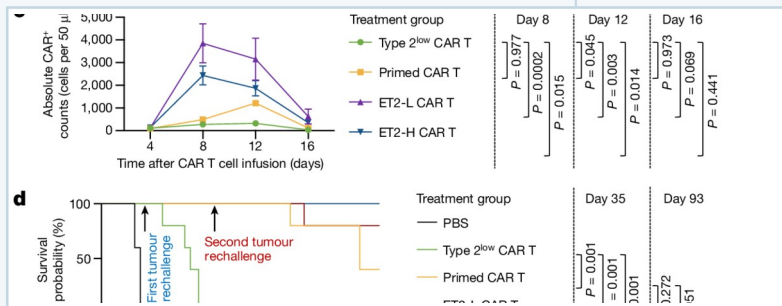


Fig. 6c-d / expansion + survival

DESIGN OPTIONS

priming: 10 ng ml⁻¹ / 12 h
ET2-L: 10 · ET2-H: 50
controlled manufacturing exposure

10
ng ml⁻¹

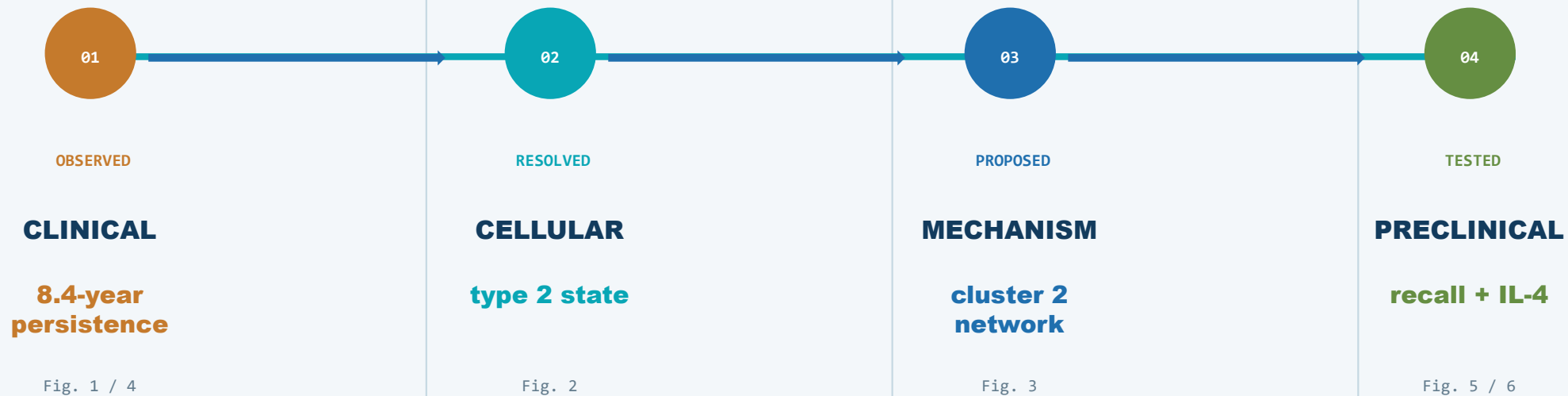
50
ng ml⁻¹

INTERPRETATION

Promising engineering lever
still bounded by model and study design

The argument is strongest when each layer stays distinct

Do not collapse clinical association, mechanism and intervention into one claim.



READING RULE

Clinical persistence anchors the story; cellular evidence resolves the state; mechanism remains a candidate; preclinical work tests the direction.

Evidence boundary: association → state definition → candidate explanation → functional test / engineering lever

THE TAKEAWAY

Type 2 function
may sustain fitness.

A durable clinical phenotype
becomes a recognizable cell state,
a candidate circuit and an engineering lever.

A strong research story shows its figures—and its
limits.

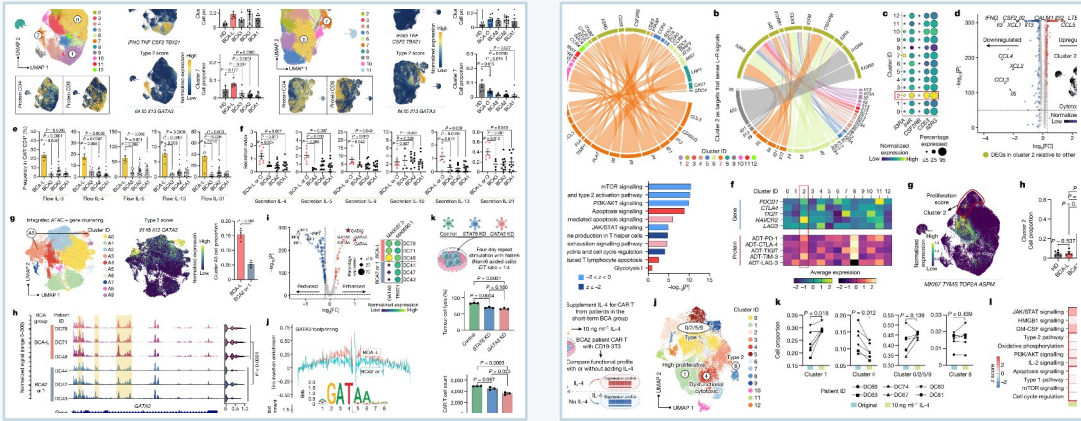


Fig. 2

Fig. 3

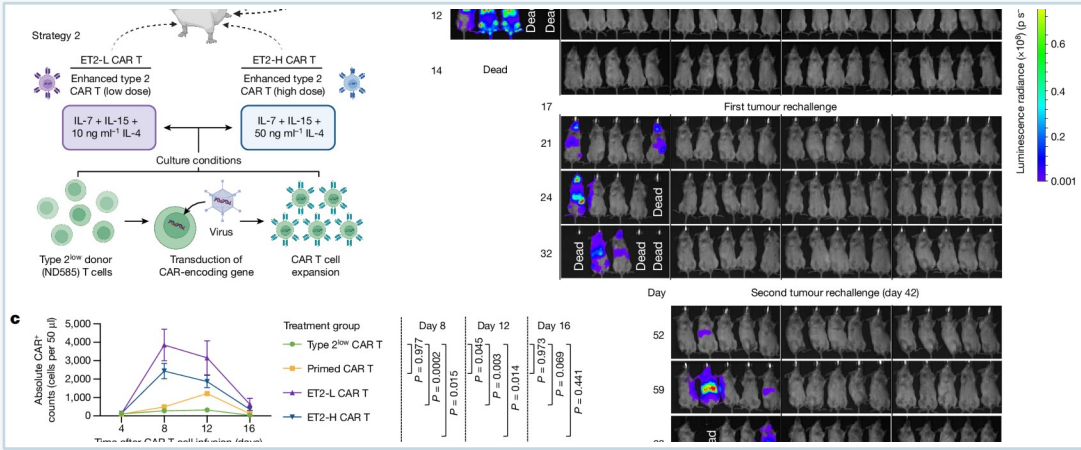


Fig. 6 / engineering strategy